week 7 lab

Pass

Why is R3 displaying a zone named “self”? What is the significance of this zone?

The self zone is present by default on all cisco routers, this zone is dedicated to any traffic that has a destination of the router or origonates from the router itself. this is nessacary for managing any traffic that is destined to the router or coming from the routers.

To test the firewall’s effectiveness, ping the follwing PCs. Where the pings successful? Explain.

PC-B from PC-A

No, we do not have any zone pairs configured with a destination of INSIDE from the INTERNET zone meaning all packets are being discarded

Ping PC-C from PC-A

No, we do not have any zone pairs configured with a destination of CONFROOM from the INTERNET zone meaning all packets are being discarded

Ping PC-A from PC-B

Yes, we have permited TCP trafic to be sent from the INSIDE security group out to the internet.

From PC-A ping R3’s G0/0/1.3 interface

Yes, because the destination of these pings are being sent to the routers self security zone which does not have a security pair confugred.

From PC-C ping R3’s G0/0/1.3 interface

Yes, because the destination of these pings are being sent to the routers self security zone which does not have a security pair confugred. Even though the CONFROOM group has been configured to explicitly forbid TCP, UDP and ICMP traffic the packets are not arriving at S0/1/1 as it is being sent directly to the router.

Credit

Create the proper zone-pair, class-maps, and policy-maps and configure R3 to prevent Internet originating traffic from reaching the Self Zone.

```
configure terminal
class-map type inspect match-any SELF_PROTOCOLS
```

```
match protocol tcp
match protocol udp
match protocol icmp
end


configure terminal
policy-map type inspect SELF_TO_INTERNET
class type inspect SELF_PROTOCOLS
inspect
end

configure terminal
zone-pair security SELF_TO_INTERNET source self destination INTERNET

zone-pair security SELF_TO_INTERNET
service-policy type inspect SELF_TO_INTERNET
```

Distinction

the following is the zone changes i have made to enable 2 different router sub interfaces communicate with each other while in the same zone

```
policy-map type inspect inside
class class-default
pass

zone-pair security inside source INSIDE destination INSIDE

zone-pair security inside
service-policy type inspect inside
```

in order to test this i had to alter some aspets of other devices

Router 3

```
int gi0/0/1.2
encapsulation dot1q 2
ip address 200.100.2.1 255.255.255.0
zone-member security INSIDE
no shut
```

Switch 3

```
vlan 2
name inside
```

```
int gi1/0/24
switchport vlan access 2
```

Router 2

```
ip route 200.100.2.0 255.255.255.0 172.16.2.1
```

with these changes I was able to confirm that PC B acting as my Vlan 3 PC and PC C acting as my VLAN 2 PC could ping each other while all other firewall properties were maintained